

Recent Developments in Modelling Drifting Interest Rates

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September 1, 2026

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Last Lecture Recap (I)

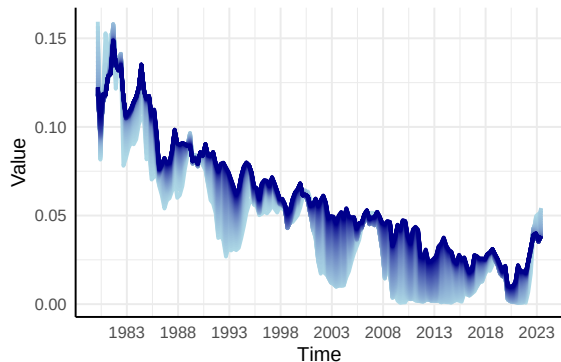


Figure: Common Trend in Yields

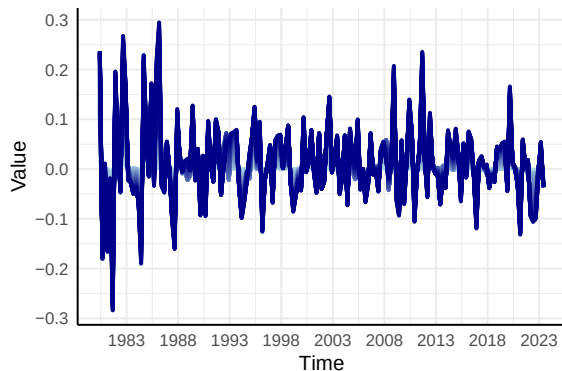


Figure: Excess Returns

- The data tells us that yields are **CO**-drifting, but excess returns are cyclical

Last Lecture Recap (II)

- AMTS models use few factors, modelled by a VAR, as the **common** drivers of the dynamics of both the expected path of future one-period rates and the term premia.
- ATMS generate term premia that are **a-cyclical** and parallel to the secular trend in yields.
- Yields are drifting, **factors are very persistent**, the forecast of the future one-period rates eventually converges to the mean of the sample-used for estimation.
- Can we do better by forecasting the drift in interest rates?

Classical A.T.S.(I)

- **Assumption:** The dynamics of a $K \times 1$ vector of state variables X_t evolve according to the following vector autoregression (VAR):

$$X_{t+1} = \mu + \Phi X_t + v_{t+1}$$
$$v_{t+1} \mid \{X_s\}_{s=0}^t \sim \mathcal{N}(0, \Sigma),$$

- In Adrian, Crump and Moench (2013), X_t consists of the first five principal components extracted from the full yield curve.
- **Assumption** The market price of risk are assumed to be in the following affine form:

$$\lambda_t = \Sigma^{-1/2} (\lambda_0 + \lambda_1 X_t),$$

- **Assumption:** No-arbitrage. That implies that there exists a pricing kernel M_t such that:

$$P_t^{(n)} = E_t \left(M_{t+1} P_{t+1}^{(n)} \right),$$

- **Assumption:** The pricing kernel, M_{t+1} , is exponentially affine:

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \Sigma^{-1/2} v_{t+1} \right),$$

- **Assumption:** $\left\{ r_{t+1}^{(n-1)}, v'_{t+1} \right\}$ are jointly normally distributed.

Why Did We Failed? - A sanity Check

- **Assumption:** The dynamics of a $K \times 1$ vector of state variables X_t evolve according to the following vector autoregression (VAR):

$$X_{t+1} = \mu + \Phi X_t + v_{t+1}$$
$$v_{t+1} \mid \{X_s\}_{s=0}^t \sim \mathcal{N}(0, \Sigma),$$

Interpreting Assumptions

In continuous time we would have:

$$dx_t = \mu_x(x_t) dt + \sigma_x(x_t) dz_t,$$

with coefficients

$$\mu_x(x) = \kappa(\bar{x} - x)$$

$$\sigma_x(x) = \Sigma s(x),$$

A Sanity Check (II)

- **Assumption:** No-arbitrage. That implies that there exists a pricing kernel M_t such that:

$$P_t^{(n)} = E_t \left(M_{t+1} P_{t+1}^{(n)} \right),$$

- **Assumption:** The pricing kernel, M_{t+1} , is exponentially affine:

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda_t' \lambda_t - \lambda_t' \Sigma^{-1/2} v_{t+1} \right),$$

Interpreting Assumptions

The assumption of N.A. guarantees that it exist a risk neutral probability $\mathcal{Q}$ such that:

$$V_t = E_t^{\mathcal{Q}} (\exp(-r_t) V_{t+1})$$

Our assumption on the pricing kernel is equivalent to assuming the pricing kernel ($\hat{\xi}_{t+1}$) is of the form:

$$\hat{\xi}_{t+1} = \hat{\xi}_t \exp \left(-\frac{1}{2} \lambda_t' \lambda_t - \lambda_t' \Sigma^{-\frac{1}{2}} v_{t+1} \right)$$

A Sanity Check (III)

Interpreting Assumptions (cont.)

- In a multivariate B-S market if we apply Girsanov we know that the radon nikodim derivative can be expressed as:

$$\frac{dQ}{dP} = \exp \left(-\frac{1}{2} \int_0^t \|\lambda(s)\|^2 ds - \int_0^t \lambda(s) dW^P(s) \right)$$

- We now prove that indeed the form of the pricing kernel is equivalent to assuming an exponentially affine Radon Nikodim:

Proof.

$$M_{t+1} = \exp(-r_t) \hat{\xi}_{t+1} / \hat{\xi}_t$$
$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \Sigma^{-\frac{1}{2}} v_{t+1} \right) \iff m_{t+1} = (-r_t) \hat{\xi}_{t+1} / \hat{\xi}_t$$



A Sanity Check (III)

- **Assumption** The market price of risk are assumed to be in the following affine form:

$$\lambda_t = \Sigma^{-1/2} (\lambda_0 + \lambda_1 X_t),$$

Interpreting Assumptions (cont.)

- In the B-S market, if all coefficients are constant, then λ is a vector of $\mathbb{R}^k$ that satisfies:

$$\mu_i - r = \sigma_i^T \lambda \text{ for } i = 1, \dots, d$$

So Where Did We fail? (I)

- (Buy-and-hold) excess returns can be written as:

$$\begin{aligned} r\mathbf{x}_{t+1}^{(n-1)} = & \underbrace{\beta^{(n-1)'} (\lambda_0 + \lambda_1 \mathbf{x}_t)}_{\text{Expected Return}} - \underbrace{\frac{1}{2} \left(\beta^{(n-1)'} \Sigma \beta^{(n-1)} + \sigma^2 \right)}_{\text{Convexity Correction}} \\ & + \underbrace{\beta^{(n-1)'} \mathbf{v}_{t+1}}_{\text{Return Innovation}} + \mathbf{e}_{t+1}^{(n-1)}. \end{aligned}$$

- In Matrix form:

$$\mathbf{r}\mathbf{x} = \left(\lambda_0 \mathbb{1}_{T \times 1}^\top + \lambda_1 \mathbf{X}_-^\top \right)^\top \mathbf{B} - \frac{1}{2} \left(\mathbf{B}^* \text{vec}(\Sigma) + \sigma^2 \mathbb{1}_{K \times 1} \right) \mathbb{1}_{T \times 1}^\top + \mathbf{V}^\top \mathbf{B} + \mathbf{E}$$

So Where Did We fail? (II)

- The model also implies that, for any maturity n ,

$$p_t^n = A_n + B_n' X_t + u_t^n.$$

- Therefore, also the risk-free rate r_t can be expressed as a linear function of the underlying factors

$$r_t = \delta_0 + \delta_1' X_t + \epsilon_t.$$

- **Problem:** Yields and excess returns are linear function of the same factors X_t .

But Why Should It Be A Problem?

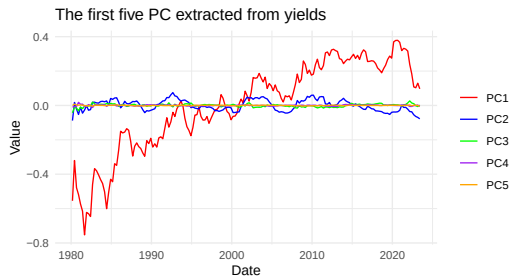


Figure: The five principal components extracted.

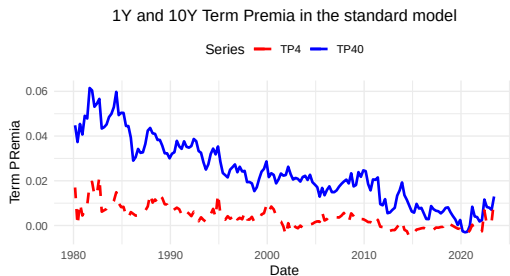


Figure: Term Premia Generated from the ATS model

- We are extracting the P.C. from non stationary data, which will likely result in one (or more) non stationary components.
- The VAR will be based on spurious regressions, high match of in sample data and bad performance O.O.S.
- How can we hope to explain a stationary series (term premium) and a non stationary one (yields) if they both depends on non stationary components?

The Favero-Fernández Fuertes Solution

- **Idea:** Use our cycle trend decomposition for the short-term rate and extend it to the entire Term Structure

$$\begin{aligned}r_t^{(1)} &= r_t^{(1),*} + u_t^{(1)} \\r_t^{(1),*} &= \gamma_1 MY_t + \gamma_2 \Delta y_t^{\text{pot}} + \gamma_3 \pi_t^{LR} \\r_t^{(n)} &= r_t^{(n),*} + u_t^{(n)} \\r_t^{(n),*} &= \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t \left[r_{t+i}^{(1),*} \right]\end{aligned}$$

- **How?:** Extract the first five principal components from the de-trended yield curve, i.e., from $\left(u_t^{(n)} \right)_{n=1}^N$.

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Explaining Falling Stars I

- To implement the proposed solution we need to *manually* extract a trend from the co-integrated yields.
- How can we explain the decline in the natural interest rate?
 - 1 Shifting in demographics due to the baby boom generation (coherent with the MY_t variable)?
 - 2 Rising Inequality?

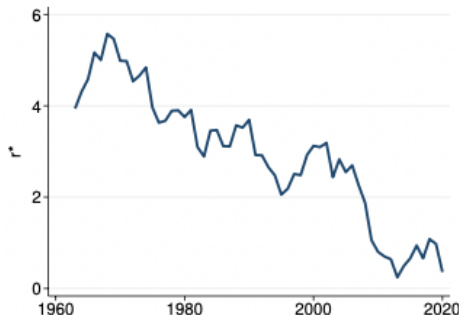


Figure: An estimate of the natural rate of interest (r^*) following Laubach and Williams (2003).

Explaining Falling Stars II

Take a simple N.K. model, the Euler equation for the representative agent in the economy is:

$$1 = \beta \mathbb{E}_t \left\{ \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} (1 + r_t) \right\}. \quad (1)$$

At steady state it must be that:

$$1 + r = \frac{\Delta C^\sigma}{\beta}$$

Or in logs:

$$\begin{aligned} r &= \sigma \Delta c - \log \beta \\ &= \sigma \Delta c - \gamma \end{aligned} \quad (2)$$

Implications

From eq. (2) Either interest rates are driven *down* by an increase in the *patience* of households or there has been a decline in the growth rate of output. Note that in our notation σ is the inverse of the I.E.S..

Where Should we Go? (I)

- Decompose the natural interest rate following *Laubach and Williams (2003)* into a the trend growth rate and other determinants:

$$r_t^* = cg_t + z_t \quad (3)$$

- Couple eq. (3) to:

$$\tilde{y}_t = a_{y,1}\tilde{y}_{t-1} + a_{y,2}\tilde{y}_{t-2} + \frac{a_r}{2} \sum_{j=1}^2 (r_{t-j} - r_{t-j}^*) + \epsilon_{1,t}, \quad (4)$$

And to:

$$\pi_t = B_\pi(L)\pi_{t-1} + b_y\tilde{y}_{t-1} + b_i(\pi'_t - \pi_t) + b_o(\pi_{t-1}^o - \pi_{t-1}) + \epsilon_{2,t}. \quad (5)$$

Laubach and Williams Procedure (I)

- Specify the transition equation for the state space model:

$$\begin{aligned}z_t &= z_{t-1} + \epsilon_{3,t} \\ y_t^* &= y_{t-1}^* + g_{t-1} + \epsilon_{4,t}, \\ g_t &= g_{t-1} + \epsilon_{5,t}.\end{aligned}\tag{6}$$

- Use the Kalman filter to estimate the state space model, but apply a 3 steps procedure to avoid the *towards 0* bias.

Laubach and Williams Procedure (II)

We would like to estimate:

$$\mathbf{y}_t = [y_t, \pi_t]'$$

$$\mathbf{x}_t = [y_{t-1}, y_{t-2}, r_{t-1}, r_{t-2}, \pi_{t-1}, \pi_{t-2}, \pi_{t-5}, \pi_{t-8}, \pi_{t-1}^0 - \pi_{t-1}, \pi_t^m - \pi_t]'$$

$$\xi_t = [y_t^*, y_{t-1}^*, y_{t-2}^*, g_{t-1}, g_{t-2}, z_{t-1}, z_{t-2}]'$$

$$\mathbf{H}' = \begin{bmatrix} 1 & -a_1 & -a_2 & -c\frac{a_3}{2} & -c\frac{a_3}{2} & \frac{-a_3}{2} & \frac{-a_3}{2} \\ 0 & -b_3 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{A}' = \begin{bmatrix} a_1 & a_2 & \frac{a_3}{2} & \frac{a_3}{2} & 0 & 0 & 0 & 0 & 0 \\ b_3 & 0 & 0 & 0 & b_1 & b_2 & 1 - b_1 - b_2 & b_4 & b_5 \end{bmatrix}$$

Laubach and Williams Procedure (III)

Remark

- In applying the Kalman filter to this model, maximum likelihood estimates of the standard deviations of the innovations to z and the trend growth rate, σ_3 and σ_5 , are likely to be biased towards 0 see Stock (1994).
- Use the Stock-Watson (1998) median unbiased estimator to obtain estimates of the ratio $\lambda_g \equiv \sigma_s/\sigma_4$, $\lambda_z \equiv (\sigma_3/\sigma_1) a_1/\sqrt{2}$.
- Apply the Kalman filter omitting the real rate gap and assuming that the growth trend rate is constant. Obtain the values for λ_g .
- Impose λ_g and starting from the previous model include also the real interest rate in the output gap assuming z is constant. Obtain an estimate of λ_z
- Impose λ_z and λ_g and estimate the full model.

Where Should We Go? (IV)

- The factor that seems to contribute the most to the decline in r^* is the "unobserved part".

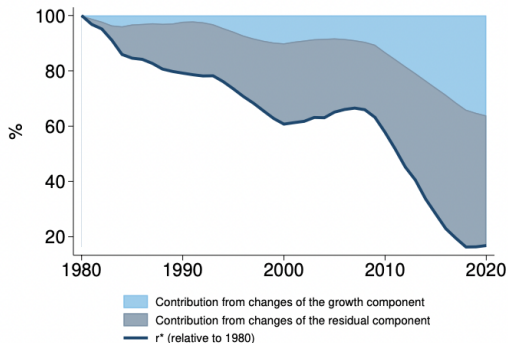


Figure: An estimate of the natural rate of interest (r^*) following Laubach and Williams (2003).

Rising Inequality and Demographic Shifts

Hypothesis (1)

Rise in inequality in lifetime income may explain an expansion in savings that pushed down r^* .

- The economic channel is simple: A shift in income towards high income households with high savings rates puts downward pressure on aggregate demand, necessitating a decline in the expected return on wealth to clear the goods market.

Hypothesis (2)

The "baby boom generation": instead of focusing on differences across the *income distribution* we focus on differences across the *age distribution*.

- Younger individuals earn nothing, and therefore must borrow from middle-age workers. The oldest workers do not save; they consume all of their wealth before dying. The borrowing by younger individuals therefore must match the saving by middle-age workers. If the cohort of middleage workers is large relative to younger individuals, interest rates must fall to clear the lendingborrowing market.

The Shift Share Methodology (I)

Let Θ_{jt} and Z_{jt} be the nominal saving and nominal income for any group j in year t , with Θ_t and Z_t being the aggregates over the groups. The change in the total saving to total income ratio for the groups from year 0 to year τ can be written as:

$$\frac{\Theta_\tau}{Z_\tau} - \frac{\Theta_0}{Z_0} = \sum_{j=1}^{j=J} (\alpha_{j\tau} - \alpha_{j0}) \cdot \frac{\Theta_{j0}}{Z_{j0}} + \sum_{j=1}^{j=J} \alpha_{j\tau} \cdot \left(\frac{\Theta_{j\tau}}{Z_{j\tau}} - \frac{\Theta_{j0}}{Z_{j0}} \right)$$

where α_{jt} is the share of income for group j in year t .

Proof.

$$\frac{\Theta_\tau}{Z_\tau} - \frac{\Theta_0}{Z_0} = \sum_{j=1}^{j=J} \alpha_{jt} \frac{\Theta_{jt}}{Z_{jt}} - \sum_{j=1}^{j=J} \alpha_{j0} \frac{\Theta_{j0}}{Z_{j0}}$$

Subtract and add from the R.H.S. side: $\sum_{j=1}^{j=J} \alpha_{jt} \frac{\Theta_{j0}}{Z_{j0}}$ regroup and the proof is completed. □

The Shift Share Methodology(II)

Let $s_{jt} = \frac{\Theta_{jt}}{Z_{jt}}$, which is the saving rate for group j out of its own income. Then the change in the total saving to total income ratio can be decomposed into a shift-share term and a residual term:

$$\frac{\Theta_{\tau}}{Z_{\tau}} - \frac{\Theta_0}{Z_0} = \underbrace{\sum_{j=1}^{j=J} (\alpha_{j\tau} - \alpha_{j0}) \cdot s_{j0}}_{\text{shift-share}} + \underbrace{\sum_{j=1}^{j=J} (s_{j\tau} - s_{j0}) \cdot \alpha_{j\tau}}_{\text{residual}} \quad (7)$$

- The first term is the critical object for empirical study. It represents the "all-else equal" prediction of what should happen to saving if there is a shift in income toward specific groups over time.
- If a certain group has a particularly high saving rate (s_j) and that group experiences a large rise in its share of income ($\alpha_{j\tau} - \alpha_{j0}$), then we can expect a large rise in saving coming from that group.

Underappreciation of Equilibrium effects?

- From eq. (7) we see that *ceteris paribus* an increase in savings from a group would tend to push down interest rates
- The macroeconomic response to a shift in saving coming from a certain group can be decomposed using the national accounting identity equating the sources and uses of saving in year t : $\Theta_{it} = I_t + F_t - \Theta_t^g - \Theta_{jt}$

Remark

A saving glut from one part of the population does not require a rise in aggregate saving even if it puts downward pressure on equilibrium interest rates.

- Suppose that we are in a closed economy in which investment is constant (for simplicity assume it to be 0) and that there are two groups in the population, rich and poor. Then it must be that $I_t = S^P + S^R \rightarrow S^P = -S^R$. Hence even with constant (aggregate savings) there is downward pressure.

Methodology I

- As dataset the authors use the SCF+ *i.e.* *Survey on Consumer Finances*
- They use the synthetic saving approach and hence they estimate nominal saving from $t - 1$ to t by group j from the following identity:

$$\Theta_{jt} = W_{jt} - W_{jt-1} \cdot (1 + \pi_t) - H_{jt}$$

where Θ_{jt} is nominal saving by group j at time t , W_{jt} is nominal wealth of group j at time t , π_t is the pure valuation gain on wealth, and H_{jt} is net inheritances going to group j at time t .

- For each survey wave households are sorted into cohorts based on the year of the household head. Each birth cohort comprises households in which the household head age spans for 10 year period. Moreover each birth cohort is braked down in to three income groups: to 10%, next 40%, bottom 50%.

Remark

Separation of income groups *within courts* allow to sterilize the effect given by age.

Empirical Results I

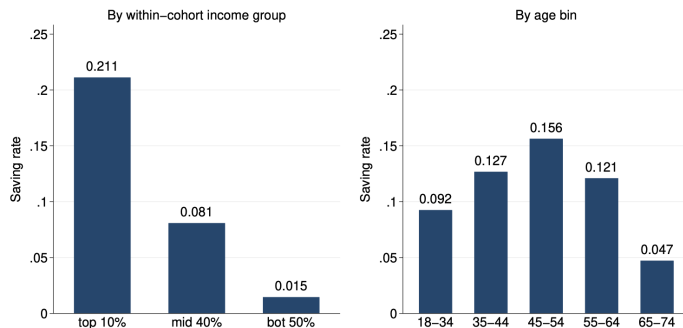


Figure: Estimated saving rates out of own income across the within-birth-cohort income distribution (left panel) and the age distribution (right panel). The sample period is 1953 to 2019 .

- The within-cohort sorting done in this study shows that this large difference is not due to life cycle effects.

Empirical Results II

Income group	Saving rate s_0	Income shift $\alpha_t - \alpha_0$	Shift-share $s_0 \cdot (\alpha_t - \alpha_0)$	Δ Saving rate $s_t - s_0$	Post income share α_t	Residual $(s_t - s_0) \cdot \alpha_t$	Total
Top 10	0.253	0.118	0.030	0.007	0.427	0.003	0.033
Next 40	0.082	-0.054	-0.004	-0.038	0.406	-0.016	-0.020
Bottom 50	0.026	-0.064	-0.002	-0.094	0.167	-0.016	-0.017

Table: This table presents the results of the shift-share methodology as outlined in equation 3 across the within-birth-cohort income distribution. Period $t = 0$ is 1963 to 1982 and period $t = \tau$ is 1995 to 2019. Annual averages are reported.

Age	Saving rate s_0	Income shift $\alpha_t - \alpha_0$	Shift-share $s_0 \cdot (\alpha_t - \alpha_0)$	Δ Saving rate $s_t - s_0$	Post income share α_t	Residual $(s_t - s_0) \cdot \alpha_t$	Total
18-34	0.076	-0.109	-0.008	-0.028	0.146	-0.004	-0.012
35-44	0.077	-0.017	-0.001	0.087	0.208	0.018	0.017
45-54	0.160	0.028	0.005	-0.013	0.253	-0.003	0.001
55-64	0.204	0.030	0.006	-0.057	0.208	-0.012	-0.006
65-74	0.101	0.039	0.004	-0.093	0.121	-0.011	-0.007

Table: This table presents the results of the shift-share methodology as outlined in equation 3 across the age distribution. Period $t = 0$ is 1963 to 1982 and period $t = \tau$ is 1995 to 2019. Annual averages are reported.

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The Main Idea

- **Idea:** Use our cycle trend decomposition for the short-term rate and extend it to the entire Term Structure

$$\begin{aligned}r_t^{(1)} &= r_t^{(1),*} + u_t^{(1)} \\ r_t^{(1),*} &= \gamma_1 MY_t + \gamma_2 \Delta y_t^{\text{pot}} + \gamma_3 \pi_t^{LR} \\ r_t^{(n)} &= r_t^{(n),*} + u_t^{(n)} \\ r_t^{(n),*} &= \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_t \left[r_{t+i}^{(1),*} \right]\end{aligned}$$

- **How?:** Extract the first five principal components from the de-trended yield curve, i.e., from $\left(u_t^{(n)} \right)_{n=1}^N$.

Modeling Drift

- They model the drift in interest rate as:

$$r_t^{(1)} = r_t^{*,(1)} + u_t^{(1)}$$

$$r_t^{*,(1)} = \gamma_1 MY_t + \gamma_2 \Delta y_t^{pot} + \gamma_3 \pi_t^{LR}$$

$$r_t^{(n)} = r_t^{*,(n)} + u_t^{(n)}$$

$$r_t^{*,(n)} = \frac{1}{n} \sum_{i=0}^{n-1} r_{t+i}^{*,(1)}$$

Remark

- The policy rate depends also on long term expectations of inflation, given that such expectations are time varying.
- If the model is correctly specified (*i.e.* if demographics, productivity and the inflation target of the central bank successfully capture the trend in nominal rates) $u_t^{(1)}$ is stationary. Since also term premia are stationary also $u_t^{(n)}$ is stationary.

Modeling Drift in Yields

Actual vs. Trend 10-Y Yield

10-Year Yield Trend

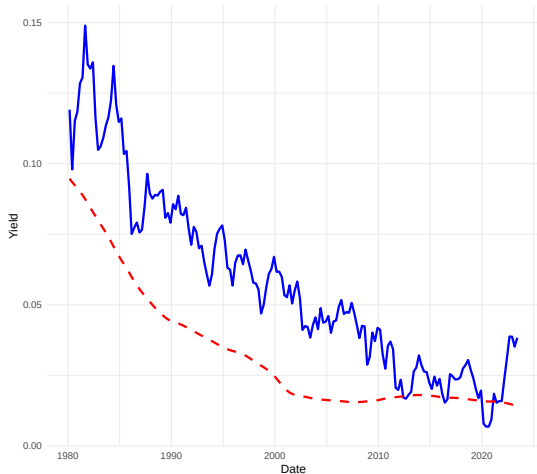


Figure: 3 months yields

Actual vs. Trend 3-m Yield

Series n_1 n_1trend

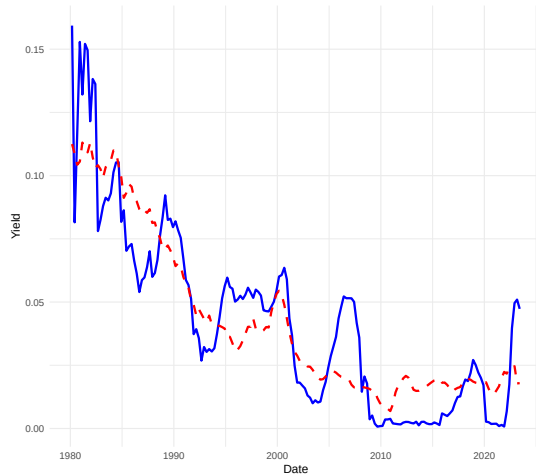


Figure: 10 year yields

Modeling Drift and Applying AMTS

- Excess returns' formulae don't change.
- Short-term rate is now written as follows

$$r_t = r_t^* + u_t$$

$$r_t^* = \gamma_1 M Y_t + \gamma_2 \Delta y_t^{\text{pot}} + \gamma_3 \pi_t^{LR}$$

$$u_t = \delta_0 + \delta_1^\top X_t$$

$$X_t = \mu + \Phi X_{t-1} + v_t$$

- The model assumptions imply that bond prices will evolve as

$$p_t^n = p_t^{n,*} + A_n + B_n' X_t + u_t^n,$$

where

$$p_t^{n,*} = p_t^{n-1,*} - \mathbb{E}_t [r_{t+n-1}^*]$$

captures the trend component of bond prices.

Classical ATS Model

- For each gross returns we have that:

$$\mathbb{E}_t(M_{t+1}R_{t+1}) = \mathbb{E}_t\left[M_{t+1}\frac{P_{t+1}}{P_t}\right] = \frac{1}{P_t}P_t = 1$$

- In the same fashion we have that:

$$P_t^{n+1} = E_t(M_{t+1}P_{t+1}^n) \tag{8}$$

Classical ATS Model (II)

We can rewrite eq. (8) as:

$$\begin{aligned} 1 &= \mathbb{E}_t \left[\exp \left\{ m_{t+1} + p_{t+1}^{(n-1)} - p_t^{(1)} \right\} \right] \\ &= \mathbb{E}_t \left[\exp \left\{ -r_t^{(1)} - \frac{1}{2} \|\lambda_t\|^2 - \lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1} + rx_{t+1}^{(n)} + r_t^{(1)} \right\} \right] \\ &= \mathbb{E}_t \left[\exp \left\{ rx_{t+1}^{(n)} - \frac{1}{2} \|\lambda_t\|^2 - \lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1} \right\} \right] \\ &= \exp \left\{ \mathbb{E}_t [\xi_{t+1}] + \frac{1}{2} \mathbb{V} [\xi_{t+1}] \right\}, \end{aligned}$$

where $\xi_{t+1} := rx_{t+1}^{(n)} - \frac{1}{2} \|\lambda_t\|^2 - \lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1}$, since $(rx_t^{(n-1)}, v_t)_{t \geq 0}$ are jointly normally distributed.

Classical ATS Model (III)

$$\begin{aligned}\mathbb{E}_t [\xi_{t+1}] &= \mathbb{E}_t \left[r x_{t+1}^{(n-1)} \right] - \frac{1}{2} \|\lambda_t\|^2 \\ \mathbb{V}_t [\xi_{t+1}] &= \mathbb{V}_t \left[r x_{t+1}^{(n-1)} - \lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1} \right] \\ &= \mathbb{V}_t \left[r x_{t+1}^{(n-1)} \right] + \mathbb{V}_t \left[\lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1} \right] - 2 \operatorname{cov} \left(r x_{t+1}^{(n-1)}, \lambda_t^T \Sigma^{-\frac{1}{2}} v_{t+1} \right) \\ &= \mathbb{V}_t \left[r x_{t+1}^{(n-1)} \right] + \lambda_t^T \Sigma^{-\frac{1}{2}} \mathbb{V}_t [v_{t+1}] \Sigma^{-\frac{1}{2}} \lambda_t - 2 \lambda_t^T \Sigma^{-\frac{1}{2}} \operatorname{cov}_t \left(r x_{t+1}^{(n-1)}, v_{t+1} \right) \\ &= \mathbb{V}_t \left[r x_{t+1}^{(n-1)} \right] + \|\lambda_t\|^2 - 2(\lambda_0 + \lambda_1 X_t) \beta_t^{(n-1)}\end{aligned}$$

Where:

$$\beta_t^{(n-1)} := \Sigma^{-1} \operatorname{cov}_t \left(r x_{t+1}^{(n-1)}, v_{t+1} \right) \in \mathbb{R}^K$$

Classical ATS Model (IV)

- Assume that $\beta_t^{(n)} = \beta^{(n)}$ for every $t \geq 0$
- Decompose the unexpected excess return, $rx_{t+1}^{(n-1)} - \mathbb{E}_t [rx_{t+1}^{(n-1)}]$ into a component that is correlated with v_{t+1} and another component which is conditionally orthogonal, $\varepsilon_{t+1}^{(n-1)}$ (return pricing error):

$$rx_{t+1}^{(n-1)} - \mathbb{E}_t [rx_{t+1}^{(n-1)}] = v_{t+1}^T \gamma^{(n-1)} + \varepsilon_{t+1}^{(n-1)}.$$

- Notice that

$$\beta_t^{(n-1)} = \Sigma^{-1} \left(\mathbb{E} [rx_{t+1}^{(n-1)} v_{t+1}] - \mathbb{E} [rx_{t+1}^{(n-1)}] \mathbb{E}_t [v_{t+1}] \right) = \Sigma^{-1} \mathbb{E} [rx_{t+1}^{(n-1)} v_{t+1}]$$

Classical ATS Model (V)

- Moreover notice that:

$$\gamma^{(n-1)} = (\mathbb{E} [v_{t+1}^T v_{t+1}])^{-1} \mathbb{E} [v_{t+1} r x_{t+1}^{(n-1)}] = \Sigma^{-1} \mathbb{E} [r x_{t+1}^{(n-1)} v_{t+1}]$$

because $\mathbb{E} [v_{t+1}^T v_{t+1}] = \Sigma$. So actually $\gamma^{(n)} = \beta^{(n)}$ for every $n \geq 0$.

Hence we can obtain:

$$\begin{aligned} \mathbb{V} [r x_{t+1}^{(n-1)}] &= \mathbb{E}_t \left[\left(r x_{t+1}^{(n-1)} - \mathbb{E}_t [r x_{t+1}^{(n-1)}] \right)^2 \right] \\ &= \mathbb{E}_t \left[\left(v_{t+1}^T \beta^{(n-1)} + \varepsilon_{t+1}^{(n-1)} \right)^2 \right] \\ &= \mathbb{E}_t \left[\left(v_{t+1}^T \beta^{(n-1)} \right)^2 + 2 v_{t+1}^T \beta^{(n-1)} \varepsilon_{t+1}^{(n-1)} + \left(\varepsilon_{t+1}^{(n-1)} \right)^2 \right] \\ &= \left(\beta^{(n-1)} \right)^T \mathbb{E}_t [v_{t+1} v_{t+1}^T] \beta^{(n-1)} + \sigma^2 \\ &= \left(\beta^{(n-1)} \right)^T \Sigma \beta^{(n-1)} + \sigma^2. \end{aligned}$$

Classical ATS Model (VI)

Theorem

We have finally obtained that:

$$rx_{t+1}^{(n-1)} = (\lambda_0 + \lambda_1 X_t)^T \beta^{(n-1)} - \frac{1}{2} \left(\left(\beta^{(n-1)} \right)^T \Sigma \beta^{(n-1)} + \sigma^2 \right) + v_{t+1}^T \beta^{(n-1)} + \varepsilon_{t+1}^{(n-1)} \quad (9)$$

- Stack eq. (9) across N maturities and T time periods to obtain:

$$\mathbf{rx} = (\lambda_0 \mathbb{1}_{T \times 1}^T + \lambda_1 \mathbf{X}_-^T)^T \mathbf{B} - \frac{1}{2} (\mathbf{B}^* \text{vec}(\Sigma) + \sigma^2 \mathbb{1}_{K \times 1}) \mathbb{1}_{T \times 1}^T + \mathbf{V}^T \mathbf{B} + \mathbf{E}$$

- $\mathbf{rx} \in \mathbb{R}^{T \times N}$.
- $\lambda_0 \in \mathbb{R}^K, \lambda_1 \in \mathbb{R}^{K \times K}$,
- $\mathbf{X}_- = [X_1 | X_2 | \dots | X_{T-1}]^T \in \mathbb{R}^{T \times K}$,
- $\mathbf{B} \in \mathbb{R}^{K \times N}$,
- $\mathbf{B}^* = [\text{vec}(B_1 B_1^T) | \dots | \text{vec}(B_n B_n^T)]^T \in \mathbb{R}^{N \times K^2}$,
- $\mathbf{V} \in \mathbb{R}^{T \times K}$ and $\mathbf{E} \in \mathbb{R}^{T \times N}$.

The Magic Trick

Table: Roots of the characteristic polynomial for the VAR specification of PC extracted from yields (Panel A) and of the VAR specification for PC extracted from the detrended yields (Panel B).

	PC1	PC2	PC3	PC4	PC5
Panel A: VAR on PCs from yields (Adrian et al., 2013)	0.9721	0.8949	0.6604	0.44	0.3306
Panel B: VAR on PCs from yields' cyclical component	0.8899	0.8899	0.7049	0.5046	0.2898

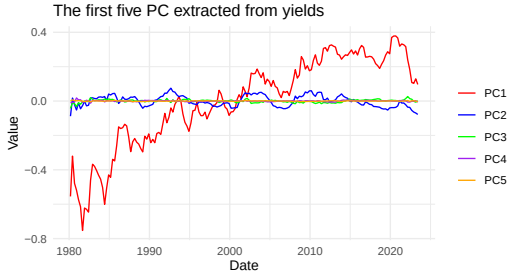


Figure: ACM model

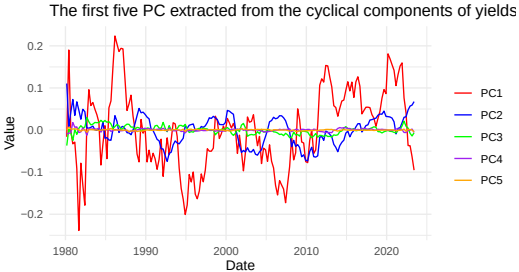


Figure: FF Model

Fit and Forecasting Performance

	n							
	4	8	12	16	20	28	40	60
Panel A: Standard ACM model								
X_t	0.17	0.13	0.10	0.09	0.09	0.09	0.10	0.11
X_t and ν_{t+1}	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
Panel B: FF								
X_t	0.15	0.14	0.13	0.13	0.13	0.13	0.14	0.13
X_t and ν_{t+1}	0.98	0.96	0.95	0.94	0.93	0.92	0.91	0.90

Table: Roots of the characteristic polynomial for the VAR specification of PC extracted from yields (Panel A) and of the VAR specification for PC extracted from the detrended yields (Panel B).

Fit and Forecasting Performance

3-month Yields

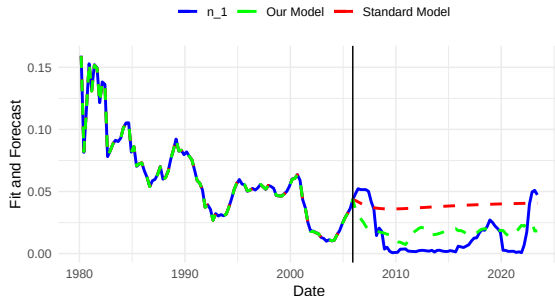


Figure: 3 months yields

10-year Yields

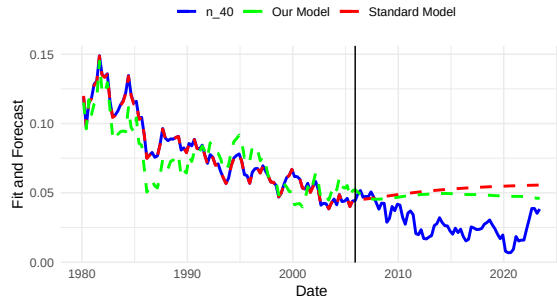


Figure: 10 year yields

Term Premia (I)

- **Definition:**

$$TP_t^{(n)} = r_t^{(n)} - \sum_{k=1}^n \mathbb{E}_t \left[r_{t+k}^{(1)} \right]$$

- If we just decompose it in $r_t^{(n)} = r_t^{*,(n)} + u_t^{(n)}$, we get

$$TP_t^{(n)} = u_t^{(n)} - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_t \left[u_{t+i}^{*,(1)} \right] + r_t^{*,(n)} - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_t \left[r_{t+i}^{*,(1)} \right]$$

- Since $r_t^{*,(n)} = \frac{1}{n} \sum_{i=1}^{n-1} r_{t+i}^{*,(1)}$, we can thus write it as

$$TP_t^{(n)} = u_t^{(n)} - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_t \left[u_{t+i}^{*,(1)} \right]$$

- **Results:** $TP^{(n)}$ has no trend.

Term Premia (II)

1Y and 10Y Term Premia in the standard model

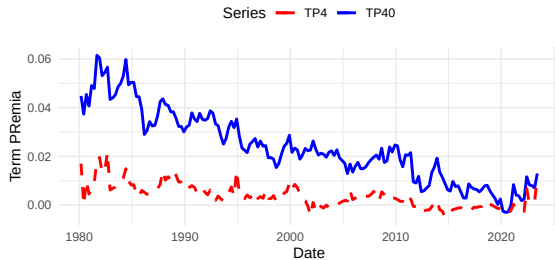


Figure: ACM model

1Y and 10Y Term Premia in our model

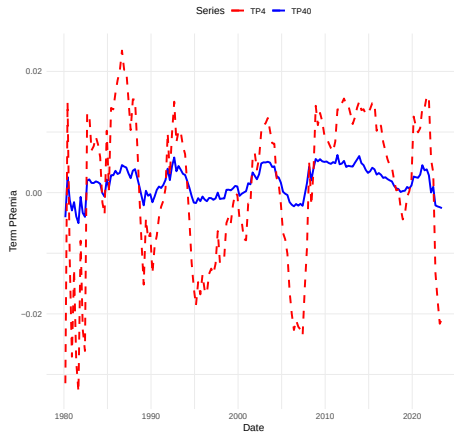


Figure: FF Model

Results of FF Approach

- Standard Affine Term Structure model do not separate trends and cycles in the data.
- The empirical model has a rather disappointing performance in predicting short-term rates and generates trending risk premia.
- FF propose a novel way to improve on the standard approach by applying the no-arbitrage restrictions to a model in which the factor structure adopted to explain holding period excess returns is extracted from de-trended yields.
- Our proposed model outperforms the standard approach in forecasting short-term rates and produces stationary risk premia, very different from those produced by the standard approach.

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A Model with Diagnostic Expectations

Paper: "*Monetary Policy and Bond Prices with Drifting Equilibrium Rates and Diagnostic Expectations*" (Favero C.A., A. Melone and A.Tamoni(2020))

Main idea:

- Bond prices are **drifting** since they reflect the drift in average expected **monetary policy rates**
- **Deviations** of bond prices from their drift should be **stationary**. They may originate from:
 - ▶ Term premia
 - ▶ Temporary **deviations from rational expectations**

Assumptions

The model is based on the following assumptions:

- The hypothesis of **Full Information Rational Expectations** (FIRE) does **not** hold.
- Equilibrium rate driven by **productivity, demographics factor, and long-term inflation expectations**
- Need to explain fluctuations in the cyclical component of yields
 - ▶ Term premia
 - ▶ Irrational expectations

Diagnostic Expectations

Diagnostic Expectations

Overreaction of agents to deviations of the monetary policy rate from its trend.

- Diagnostic expectations explain between 4% and 33% of the variability in the cyclical components of yields
- The importance of diagnostic expectations declines with the maturity of the bond

	$u_t^{(8)}$	$u_t^{(12)}$	$u_t^{(20)}$	$u_t^{(28)}$	$u_t^{(40)}$
	(1)	(2)	(3)	(4)	(5)
$(y_t^{(1)} - y_t^*)$	0.379*** (0.058)	0.337*** (0.059)	0.248*** (0.064)	0.189*** (0.065)	0.141** (0.068)
Observations	160	160	160	160	160
R ²	0.331	0.257	0.140	0.080	0.043
Implied θ	0.42	0.41	0.37	0.34	0.32

Figure: Testing Diagnostic Expectations ¹

¹The table reports OLS estimates for the regression $u_t^{(n)} = \beta (y_t^{(1)} - y_t^*) + \epsilon_t$

Bond yields

Bond yields can be decomposed into formula

- a drifting component $\left(y_t^{(n)*}\right)$
- a residual cyclical component, i.e. deviation from the drift, $\left(\delta_0 + u_t^{(n)}\right)$.

What is the **economic interpretation**?

- ▶ Presence of diagnostic expectations
- ▶ Term premium

Modeling Monetary Policy (I)

Monetary policy rules specify the dynamics of the **short-term rate** $y_t^{(1)}$:

$$\begin{aligned}y_t^{(1)} &= y_t^* + \beta' X_t + u_t^{(1)} \\ u_t^{(1)} &= \rho u_{t-1}^{(1)} + \epsilon_t^{(1)}\end{aligned}$$

where

- y_t^* is the equilibrium monetary policy rate
- X_t is a vector of stationary monetary policy drivers
- ρ is monetary policy persistence.

Modeling Monetary Policy (II)

"Cointegrating" approach to drifting policy rates, where y_t^* is the **stochastic trend** that drives drifting policy rates:

$$y_t^{(1)} = y_t^* + \beta_1 E_t(\pi_{t+1} - \pi_{t+1}^*) + \beta_2 E_t(x_{t+1}) + u_t^{(1)}$$

$$y_t^* = \gamma_1 MY_t + \gamma_2 \Delta x_t^{pot} + \gamma_3 \pi_t^*$$

$$u_t^{(1)} = \rho_1 u_{t-1}^{(1)} + \epsilon_t^{(1)}$$

MY_t , Δx_t^{pot} and π_t^* are non-stationary, they represent the **drivers of the drifting equilibrium rate**.

- $y_t^{(1)}$ is the one-period yield
- y_t^* is the equilibrium nominal rate
- π_t change in Personal Consumption Expenditures
- π_t^* is the Fed perceived target rate
- MY_t is ratio of middle-aged to young population
- Δx_t^{pot} potential output growth

Comparison between cointegrated model and restricted model (I)

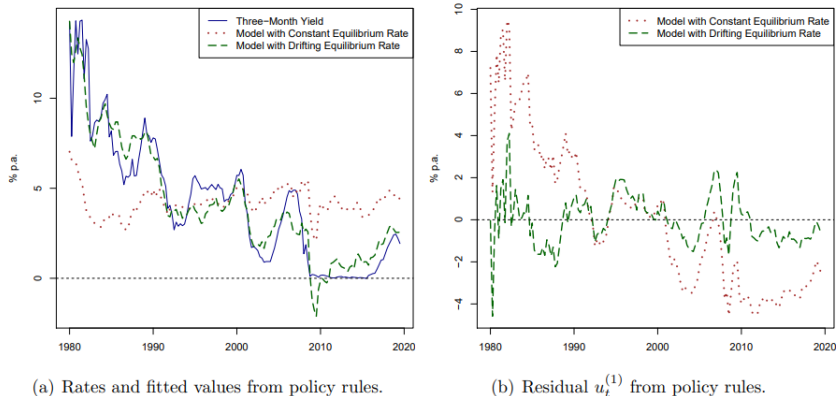


Figure: Actual vs Fitted Short-Term Rate

The model of the monetary rule with a **drifting equilibrium rate** tracks well the short-term rate movements throughout the sample.

Comparison between cointegrated model and restricted model (II)

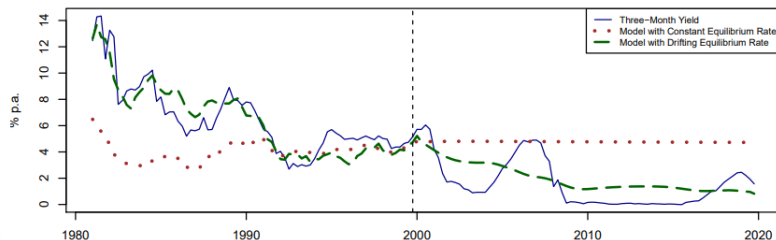


Figure: Short-Term Rate Forecasts

The drifting monetary policy rule tracks well the future evolution of the short rate for the **out-of-sample period**.

Modeling a Drifting Term Structure

The *drift in monetary policy rates* is used to model the *drift in the entire term structure*.

Yields at all maturities are decomposed into a **trend**, $y_t^{(n)*}$, and a **cyclical component**, $\delta_0 + u_t^{(n)}$.

$$\begin{aligned}y_t^{(n)} &= y_t^{(n)*} + \delta_0 + u_t^{(n)} \\u_t^{(n)} &= \rho_n u_{t-1}^{(n)} + \epsilon_t^{(n)} \\y_t^{(n)*} &= \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t[y_{t+i}^{(1)}].\end{aligned}$$

- The **trend** is the **average of expected monetary policy rates** over the duration of the bond
- the **cyclical component** is the stationary **residuals** from the $(1, -1)$ **cointegrating relationship** between **yields and their drift**.

Misspecification test for term structure models

The validity of the model with drifting monetary policy rates and bond prices is tested by checking the **existence of cointegrating** relationships with parameters $(1, -1)$ between $y_t^{(n)}$ and $y_t^{(n)*}$.

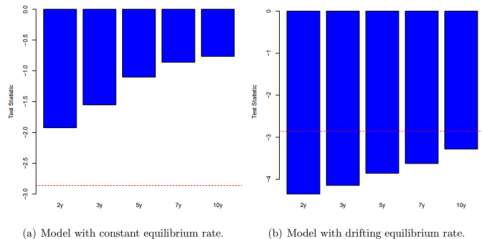


Figure: Engle and Granger (1987) Cointegration Test

Evidence to **reject** the null hypothesis of **absence of cointegrating** relationship for all maturities.

Diagnostic expectations

A stationary $u_t^{(n)}$ is **consistent with temporary deviations from Rational Expectations** generated within a Diagnostic Expectations framework (see Gennaioli and Shleifer (2018)), where long rates over-react relative to change in expectations about short rates.

Diagnostic Expectations Hypothesis

Under the diagnostic expectations hypothesis agents **over-react** to the stationary **deviations of monetary policy from its trend**:

$$E^D[y_{t+i}^{(1)}|I_t] = E[y_{t+i}^{(1)}|I_t] + \theta(E[y_{t+i}^{(1)}|I_t] - E[y_{t+i}^*|I_t]).$$

New specification for the drift in yields:

$$y_t^{(n)*} = \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t^D[y_{t+i}^{(1)}]$$

Modeling a Drifting Term Structure with Diagnostic Expectations

Hence, the **stationary component** $u_t^{(n)}$ can in principle be **explained** by the over-reaction induced by **diagnostic expectation**:

$$y_t^{(n)} = \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t[y_{t+i}^{(1)}] + \delta_0 + \underbrace{\left(\frac{1}{n}\right) \sum_{i=0}^{n-1} (E_t^D[y_{t+i}^{(1)}] - E_t[y_{t+i}^{(1)}])}_{u_t^{(n)}}$$

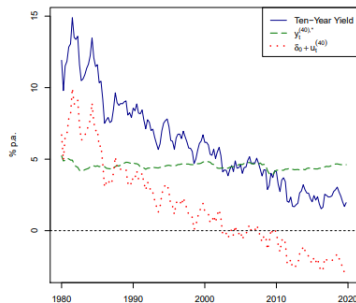
It can be derived that:

$$u_t^{(n)} = \left(\frac{1}{n}\right) \theta \frac{1 - \phi^n}{1 - \phi} (y_t^{(1)} - y_t^*)$$

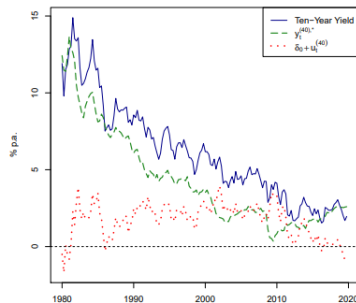
Deviations from rational expectations depend on the **parameter** θ and on the **persistence of the deviations** of monetary policy rates from the trend. At long horizons, diagnostic expectations for the monetary policy rates will converge towards rational expectations.

The dynamics of cyclical yields components (I)

What is the behavior of the residual $u_t^{(n)}$?



(a) Model with constant equilibrium rates.



(b) Model with drifting equilibrium rates.

Figure: Decomposing long-term rates

The residuals follow a random walk under the classical model with constant equilibrium rates, but are **stationary** in the model with drifting rates.

The dynamics of cyclical yields components (II)

Importantly, the estimated **deviations of bond prices** from their drift **comove** strongly with state-of-the-art **term premium estimates**:

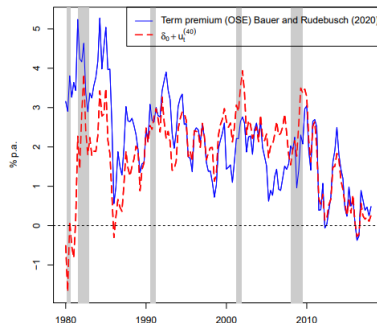


Figure: Cyclical component from model with drifting equilibrium rates vs. term premium estimate

Therefore, a fraction of what is deemed to be risk premium is a mere reflection of temporary deviations from **rational expectations**.

Trend-cycle yield decomposition

Finally, long-term interest rates depend on three components:

$$y_t^{(n)} = \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t[y_{t+i}^*] + \left(\frac{1}{n}\right) \sum_{i=0}^{n-1} E_t[y_{t+i}^{(1)} - y_{t+i}^*] + \delta_0 + u_t^{(n)}$$

- ➊ Trend component reflecting the **drift** in short-term rates
- ➋ First cyclical component related to **fluctuations of short-term rates** around their time-varying mean
- ➌ Second cyclical component related to **term premia** and possibly **deviations from rational expectations**

Having **not imposed no-arbitrage** restrictions, the cyclical component is compatible with investors **overreacting** to deviations of policy rates from its trend leading to **overestimation of future short rates** and lower bond returns.

Predicting Holding Period Excess Returns (I)

Predictability of yields also implies **predictability of holding period excess returns** on the basis of the **stationary deviations** of bond yields from their drift.

The **expected excess returns** from holding for one period the n-period bond can be written as:

$$\begin{aligned} E_t(r_{t+1}^{(n)}) &= y_t^{(n)} n - (n-1)E_t(y_{t+1}^{(n-1)}) - y_t^{(1)} \\ &= y_t^{(n)} - y_t^{(1)} - (n-1)\left(E_t(y_{t+1}^{(n-1)}) - y_t^{(n-1)}\right) - (n-1)\left(y_t^{(n-1)} - y_t^{(n)}\right)^2 \end{aligned}$$

What is new? **Forecastability** of the level of the term structure, $E_t(y_{t+1}^{(n-1)}) - y_t^{(n-1)}$.

²where $y_t^{(n)} - y_t^{(1)}$ is the slope of the term structure and $(E_t(y_{t+1}^{(n-1)}) - y_t^{(n-1)})$ is the expected change in prices of the (n-1)-maturity bond. ▶

Predicting Holding Period Excess Returns (II)

Persistent but **stationary deviations** of bond prices from their drift, $u_t^{(n-1)}$, show up as a natural **predictor of excess bond returns**:

$$E_t(y_{t+1}^{(n-1)}) - y_t^{(n-1)} = E_t\left(y_{t+1}^{(n-1)*} - y_t^{(n-1)*}\right) + (\rho_{(n-1)} - 1) \underbrace{\left(y_t^{(n-1)} - y_t^{(n-1)*} - \delta_0\right)}_{u_t^{(n-1)}}$$

Thank you!